Patricia Guerra Balboa

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LinkedIn Google Scholar ORCID

Education

PhD in Computer Science October 2021 – May 2026
Karlsruhe Institute of Technology Karlsruhe, Germany

Master of Science in Mathematics Sep. 2020 – July 2021
University of Santiago de Compostela Santiago de Compostela, Spain

Bachelor's Degree in Mathematics Sep. 2016 – July 2020
University of Santiago de Compostela Santiago de Compostela, Spain

Work Experience

Postdoctoral researcher May 2026 – Nowadays
Chair of Privacy and Security, Karlsruhe Institute for technology Karlsruhe, Germany

- Research on privacy enhancing technologies with focus on differential privacy and inference attacks.

Doctoral researcher October 2021 – May 2026
Chair of Privacy and Security, Karlsruhe Institute for technology Karlsruhe, Germany

- Formal Study of Differential Privacy in Complex and Correlated Data.

High-level research technician July 2021 – September 2022
Universitat Politécnica de Catalunya Barcelona, Spain

- Short-term research project about Generating Synthetic Data with Differential Privacy Guarantees.

High-level research technician March 2021 – July 2021
University of Santiago de Compostela Santiago de Compostela, Spain

- Studying and analysis of localization and cohomology on quasicoherent sheaves on schemes and associated spaces. Presentation about schemes package on Lean library.

Intern July 2019 – August 2019
Kidcode S.L. Vigo, Spain

- Scientific communicator and teacher.

Honors and Awards

Audience Award & Silver Medal in the FameLab Germany – Regional Heat 2024
FameLab Germany—Science Communication Competition

Best Poster Award at the Community Congress by Startup Secure 2023
K.I.T. Gründer Schmiede

"Mellores expedientes"(Best records) Award 2016
University of Santiago de Compostela

Publications

[1] P. Guerra-Balboa, A. Sauer, H. Arcolezi, and T. Strufe. "Understanding Disclosure Risk in Differential Privacy with Applications to Noise Calibration and Auditing". In: *Proceedings of the VLDB Endowment* (2026). DOI: 10.14778/3801059.3801069.

[2] M. Lange, P. Guerra-Balboa, J. Parra-Arnau, and T. Strufe. "Balancing Privacy and Utility in Correlated Data: A Study of Bayesian Differential Privacy". In: *Proceedings of the VLDB Endowment* 18.11 (2025). DOI: 10.14778/3749646.3749679.

[3] P. Guerra-Balboa, À. Miranda-Pascual, J. Parra-Arnau, and T. Strufe. "Composition in Differential Privacy for General Granularity Notions". In: *37th IEEE Computer Security Foundations Symposium (CSF)* (2024). DOI: 10.1109/CSF61375.2024.00004.

[4] P. Guerra-Balboa, A. Sauer, and T. Strufe. "Analysis and Measurement of Attack Resilience of Differential Privacy". In: *Proceedings of the 23rd Workshop on Privacy in the Electronic Society*. WPES '24. ACM, 2024. ISBN: 9798400712395. DOI: 10.1145/3689943.3695046.

[5] À. Miranda-Pascual, P. Guerra-Balboa, J. Parra-Arnau, J. Forné, and T. Strufe. "An overview of proposals towards the privacy-preserving publication of trajectory data". In: *International Journal of Information Security (IJIS)* (2024). DOI: 10.1007/s10207-024-00894-0.

[6] À. Miranda-Pascual, P. Guerra-Balboa, J. Parra-Arnau, J. Forné, and T. Strufe. "SoK: Differentially Private Publication of Trajectory Data". In: *Proceedings on Privacy Enhancing Technologies (PoPETs)* (2023). DOI: 10.56553/popets-2023-0065.

[7] P. Guerra-Balboa, A. M. Pascual, J. Parra-Arnau, J. Forné, and T. Strufe. "Anonymizing Trajectory Data: Limitations and Opportunities". In: *The Third AAAI Workshop on Privacy-Preserving Artificial Intelligence (PPAI-22)* (2022). URL: https://aaai-ppai22.github.io/files/25.pdf.

Invited talks and Presentations

Invited talks

"ε? Connecting Differential Privacy to Real-World Attack Mitigation." — May 2026
Invited talk in the Max Planck Institute for Security and Privacy — Ruhr University Bochum

"DP Noise Calibration for Correlation-Resilient Privacy Guarantees" — February 2026
Invited talk in the ANYMOS Seminar — University of Lübeck

"Balancing Privacy and Utility in Correlated Data" — November 2025
Invited talk in the PreMeDICal Seminar — INRIA Montpellier

"¿Cómo pueden las matemáticas proteger tu privacidad?" — December 2024
Peregrinando: International carrers in mathematics and physics workshop — CITMAga & IGFAE

"Unlocking the potential of composition in DP" — October 2023
Invited talk in the COSIC Seminar — KU Leuven

"Mathematics behind privacy" — June 2022
Inivited talk in García Rodeja Seminar (CITMAGA) — University of Santiago de Compostela

"How sheaves theory can help air-controllers at an airport" — March 2021
Invited talk in S.I.I. (Seminario de iniciación a la investigación) — University of Santiago de Compostela

Presentations at International Conferences

"Understanding Disclosure Risk in Differential Privacy" — September 2026
52st International Conference on Very Large Data Bases (VLDB 2026) — Boston, USA

"Balancing Privacy and Utility in Correlated Data" — September 2025
51st International Conference on Very Large Data Bases (VLDB 2025) — London, UK

"Analysis and Measurement of Attack Resilience of Differential Privacy" — October 2024
23rd Workshop on Privacy in the Electronic Society (WPES 2024) — Salt Lake City, USA

"Composition in Differential Privacy for General Granularity Notions" — July 2024
37th IEEE Computer Security Foundations Symposium (CSF 2024) — Enschede, Netherlands

"Anonymizing Trajectory Data: Limitations and Opportunities" — February 2022
Third AAAI Workshop on Privacy-Preserving Artificial Intelligence (PPAI 2022) — Online

Presentations at National Conferences

"DP to BDP: Noise Recalibration for Correlation-Resilient Privacy Guarantees" — March 2026
XIX Spanish Meeting on Cryptography and Information Security (RECSI26) — University of La Laguna

"Publishing Trajectories: A Study on Privacy Protection" — October 2022
XVII Spanish Meeting on Cryptography and Information Security (RECSI22) — University of Cantabria

Other Research Activities

PC member in ACM Conference on Computer and Communications Security (CCS 2026) and Workshop on Privacy in the Electronic Society (WPES 2026).

Reviewer for ACM Conference on Computer and Communication Security (CCS), IEEE Transactions on Information Forensics and Security (IEEE TIFS), ACM ASIA Conference on Computer and Communication Security (ASIACCS), European Symposium on Research in Computer Security (ESORICS).

Organizer of PET-CON 2026.1: 15th Privacy Enhancing Techniques Convention.

Associated Researcher on the EU project PROPOLIS – Privacy for Smart Cities (https://propolis-project.eu/).

Specific Formation

Seminar PETs and AI: Privacy Washing and the Need for a PETs Evaluation Framework 2025
Schloss Dagstuhl – Leibniz-Zentrum für Informatik (LZI) Wadern, Germany

Seminar in Scientific Communication 2023
National Institute of Science Communication (NaWik) Karlsruhe, Germany

5th Interdisciplinary Summerschool on Privacy (ISP 2023) 2023
Radboud University Berg en Dal, the Netherlands

Seminar in Schemes Theory (Algebraic Geometry) 2021
University of Santiago de Compostela Santiago de Compostela, Spain

Discrete Morse Theory & **Introduction to Ergodic Theory course** 2019
R.E.T. In collaboration with U.S.C Santiago de Compostela, Spain

Course on scientific documents and presentations on LaTex 2019
University of Santiago de Compostela Santiago de Compostela, Spain

Mechanical computer assisted design Campus 2014
Carlos III de Madrid University Madrid, Spain

Teaching Experience

Lecturer, Privacy Enhancing Technologies Every Summer Semester, 2022–present
Master's course Karlsruhe Institute of Technology (KIT)

Organizer, Seminar on Privacy and Technical Data Protection Every Summer Semester, 2022–present
Master's course in IT Security Karlsruhe Institute of Technology (KIT)

Organizer, Seminar on Privacy and Security Every Winter Semester, 2023–present
Master's course in IT Security Karlsruhe Institute of Technology (KIT)

Teaching Assistant, Reading Group on Privacy Enhancing Technologies Summer Semester 2022
Master's course in IT Security Karlsruhe Institute of Technology (KIT)

Supervised Bachelor's and Master's Theses

Measurement of Attack Resilience of Differential Privacy 2025
Master's Thesis KIT

Utility of Semantic Privacy Notions for Correlated Data 2024
Master's Thesis KIT & UPC

Analysis of Topological Privacy Notions 2024
Master's Thesis KIT

LDP protection of trajectories based on dependency reduction 2024
Bachelor's Thesis KIT

Riemannian-based Clustering for Differential Private Location Publication 2023
Master's Thesis KIT

Privacy Analysis of Deep Graph Generative Models using M.I.A.s 2023
Master's Thesis KIT

Languages

Spanish: Native — English: Fluent — Galician: Fluent — German: Intermediate